Mars Terraforming — Gameplay Guide

What this game is

You are in charge of a colony on **Mars**. You begin with a small landing capsule, a handful of colonists and a few supplies — and your mission is enormous: to turn a frozen, lifeless planet into a world where people can walk outside without a suit.

It is a real-time **strategy and management** game (pausable at any moment). You put up buildings, manage resources, look after your crew, research new technologies, and gradually warm and awaken an entire planet. There are no rival players — your real opponent is Mars itself: the cold, the thin air, the radiation and the scarcity.

The big goal: terraforming Mars

At the top of the screen sit **four target gauges**. To win, all four must reach **100%**:

- 🌡️ **Temperature** — you warm the planet with greenhouse-gas factories, vast orbital mirrors that focus sunlight, and so on.
- 🎈 **Pressure** — you thicken the atmosphere so that it can hold onto heat and air.
- 💨 **Oxygen** — you fill the air with breathable oxygen, mostly through plants and microorganisms.
- 💧 **Water** — you melt the ice caps and drop comets until liquid water covers the surface.

When all four hit 100% — **and your crew is still alive** — you have won: Mars is habitable.

100% is the target, not a score to beat. Once a gauge passes its target it stops showing a percentage and starts showing **how far over** you are (e.g. `+16.0 C`) with an up-arrow: amber means the extra output is mostly wasted, **red means danger** — temperature or pressure more than 4 over target is runaway-greenhouse territory, where the crew sickens, vegetation withers and the oceans boil away. A line at the bottom of the screen spells it out. Throttle or dismantle greenhouse-gas factories and mirrors before it gets there.

You lose if the colony collapses: if life support runs dry and your colonists have no air, water or food left.

It is a long journey: you start by keeping a few people alive, and you end up reshaping an entire planet.

> 🌍 And then?
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> Winning **is not the end of the game**. The moment Mars becomes habitable, **Phase 2: The Living Planet** begins — an entire second half, in which the world you built starts to have a life, a set of problems, and a will of its own. See below.

The planet and the map

You play on a map of **hexagons**. Each hex has:

- **Terrain** — plains, lowlands, highlands, craters. As you terraform, **water** appears, and later **vegetation**.
- **Elevation** — it decides where you can build and where the water will flood.
- **Deposits** — buried treasure, shown as coloured diamonds:
  - **Ice (H_2O)** → water
  - **Iron (Fe)** → materials
  - **Silicon (Si)** → electronics & exports
  - **Regolith (Rg)** → raw feedstock for materials

You can pan and zoom the camera to scout the map and find the best spots for your mines.

Your resources

The colony runs on a resource "economy". The bar at the top shows how much of each you hold — and the numbers turn **red when they are falling**, to warn you.

- ⚡ **Energy** — the "power" behind every building. If it runs out, you get a blackout and things stop working.
- 💧 **Water** — for the crew and for growing food.
- 💨 **Oxygen** — for your colonists to breathe (separate from the atmospheric-oxygen target).
- 🍎 **Food** — to feed the crew.
- 📦 **Materials** — the "bricks and metal" that buildings are made of.
- 🔷 **Silicon** — for advanced buildings, and to sell to Earth.
- 💰 **Credits** — money; you spend it to build and earn it from exports.
- 👷 **Crew** — your people (see below).

The landing capsule is the central **warehouse**: everything is pooled there. Every building either **produces** or **consumes** resources every moment, so the trick is to keep the balance positive.

The crew (your colonists)

Colonists are precious. Each has a **specialty**, and putting them in the "right" building makes it perform better:

- **Geologist** — better at mines & drills, spots hidden deposits.
- **Engineer** — faster construction, better factories, fewer breakdowns.
- **Botanist** — more food and faster-spreading vegetation.
- **Climatologist** — a bonus to the great planetary-engineering works.
- **Doctor** — appears in **Phase 2**; staffs the Isolation Hospitals and is your only defence against the **Martian Plague**.

How you manage them: right-click a building to select it, then use the **[+] / [–]** buttons (or the +/– keys) to assign or remove workers. A staffed building produces far more than an empty one.

Repairs: a building knocked out by an event, a marsquake or a storm stops producing until it is repaired. In the crew screen those buildings sit **at the top**, in red, with a **repair slot** — and that includes buildings that normally run themselves, which have no jobs at all. Drop a colonist in it and the repair speeds up: an **Engineer repairs three times faster** than nobody, anyone else twice. The moment the repair finishes, that colonist is released back into the unassigned pool on their own.

The crew screen (C): press **C** — or the two-colonists icon on the bottom bar — to see every colonist and every job in one place: the unassigned pool on the left, all the buildings with jobs on the right. **Drag** a colonist onto a slot to move them there; if that slot is already taken, the two colonists **swap buildings**. Drop someone on *UNASSIGNED* to pull them off the job. A `★` marks a colonist whose specialty matches the building (+50% output), and each building shows its current efficiency, so you can see at a glance where a reshuffle is worth it.

The population **grows over time**, as long as there is housing and living conditions (food, health, morale) are good.

The buildings

You build by clicking the buildings icon on the bottom bar, picking a building, then clicking a hex. Every building costs **Credits up front**, plus **Materials while it is being built**. Mines must be placed on top of the right deposit.

The main categories:

- 🏠 **Housing** — the landing capsule, habitat modules, and later whole "domeless cities" for a huge population once the air is breathable.
- ⚡ **Power** — solar panels (which fade in dust storms), batteries (storage), and fission reactors (steady power, independent of the sun).
- 🫧 **Life Support** — oxygen recyclers, ice drills for water.
- 🌱 **Food** — hydroponics bays.
- 🏭 **Industry** — regolith printers, iron/silicon mines, and export terminals (which sell silicon to Earth for steady income).
- 🔬 **Research** — laboratories that generate research points.
- 🪐 **Planetary Engineering** (the "big guns" of terraforming) — greenhouse-gas factories, orbital mirrors, an artificial magnetosphere station (which protects the atmosphere), comet redirectors (which bring water & pressure).
- 🌳 **Biosphere** — cyanobacteria, genetically modified forests, wildlife reserves; they produce oxygen and **spread vegetation** across the surface.

Research & Technology

Your laboratories generate **research points**. You open the research menu, choose a technology, and wait for enough points to accumulate to unlock it. New technologies unlock **new buildings** and abilities.

Technology advances in **tiers**:

1. **Survival** — you know this from the start (the basic buildings).
2. **Industrial Expansion** — fission power, metallurgy, silicon extraction, exports, greenhouse-gas factories.
3. **Planetary Engineering** — orbital mirrors, artificial magnetosphere, comet redirection.
4. **Biosphere & Ecosystem** — microorganisms, hardy trees, and finally open-air cities.

Plus two more tiers that **unlock only after you have terraformed the planet** (Phase 2):

5. **Society & Climate** — carbon capture, political governance, pollution scrubbing, macro-epidemiology.
6. **Infrastructure & Automation** — quantum processors, mantle drilling, drone swarms, storm hardening, gene banks, maglev vacuum trains.

Every technology has **prerequisites**, so you climb a tree from the basics towards the spectacular. When a piece of research completes, a window tells you what has just become available.

Reclaim

A special technology lets you **dismantle buildings** and get back part of their Credits and Materials — handy for tearing down a mine that has exhausted its deposit and putting those resources to work elsewhere.

Opening up new ground

Two technologies do not unlock a building — they unlock **terrain you could not build on before**:

- **Mountain Foundations** (Phase 2, after *Heavy Metallurgy*) — rock anchors and terraced platforms. Mountains stop being off-limits, which matters because the iron and silicon veins run right through them.
- **Offshore Drill Platforms** (Phase 3, after *Orbital Mirrors*) — once the mirrors melt the caps, the polar ice becomes sea, but the ice deposit is still down there. Floating rigs let you put an **ice drill on water that hides ice** — everything else still needs dry land.

Buildings that need a specific deposit still need it, and the build cursor tells you which research is missing (e.g. "*Mountain: needs Mountain Foundations*").

Time is in your hands

You control the pace of the simulation:

- **Pause** — to think it through in peace.
- **×1 / ×2 / ×4** — normal, fast, very fast, for when you are simply waiting for the planet to "cook".

Terraforming is a slow process, so you will often speed things up and step in only when needed.

Events & threats

Mars does not leave you alone. **Random events** strike from time to time (announced in a popup):

- 🌪️ **Dust storm** — solar output collapses; if you rely on solar alone, a blackout is coming.
- 🌞 **Solar flare** — radiation that sickens colonists and fries electronics, unless you have a **magnetosphere** or a natural **cave shelter**.

- 🔧 **Life-support failure** — an oxygen/water system breaks and needs an **Engineer** to repair it.
- 🕳 **Cave discovery** — (a good one!) natural shielding against radiation.

Alongside those, you must watch the standing dangers: low **energy** (brownout), falling **crew health**, and **atmospheric loss** (which stops once you build a magnetosphere). Warnings appear along the bottom of the screen, together with **how many buildings need crew** or **have run out of deposit** — so you can rush to wherever you are needed.

In **Phase 2** that same warning line fills up with new, nastier messages: *runaway greenhouse*, *stagnation*, *strike*, *heavy pollution*, *imminent marsquake*, *super-storm*, *invasive species*, *plague*, *logistics blackout*. Each one tells you exactly **what to build** to stop it — if you read it in time.

🌍 Phase 2: The Living Planet

The moment all four gauges hit 100%, the game **doesn't end** — it changes character.

Until now Mars was a dead rock you were pushing around. Now it is a **living planet**: it has an atmosphere that moves, seas that swell, ecosystems that evolve, and — most dangerous of all — **tens of thousands of people with opinions of their own**.

Two things change immediately:

- **The population explodes.** You are no longer counting a dozen colonists by name. Ships from Earth pour **thousands** of migrants onto a planet that is finally habitable — and the number climbs into the **tens of thousands**. A new population counter appears on the HUD (residents / housing).
- **Two new tech tiers and 12 new buildings unlock**, built for one purpose: to deal with the problems of a world that is now alive.

And the problems do come. Each has an answer — but you have to see it coming.

The crises of Phase 2

🔥 The greenhouse runs away

The gas factories and mirrors that **won you the game** don't switch themselves off. Temperature and pressure keep climbing past the target, the **oceans begin to boil away**, and the crew falls ill.

> **The answer:** the **Cryo-Carbon Capturer** sucks the excess CO_2 out of the air and freezes it into dry ice. Or just tear down a few factories — your call.

🌆 The Great Migration

Waves of migrants arrive without warning. If the population outgrows your **housing**, society falls into **stagnation**: people are packed in, morale drops, and production across the **entire** colony grinds down.

> **The answer: High-Density Arcologies** — vertical megacities that house tens of thousands.

⚖️ Politics: Industrialists vs Ecologists

Your residents split into two factions. The **Industrialists** want mines, factories, growth. The **Ecologists** want clean air and forests. Whatever pleases one enrages the other — and pollution enrages the Ecologists especially. If a faction's approval falls too low, it goes on **strike**: your mines and factories (or your entire biosphere) simply **stop working**.

> **The answer:** the **District Town Hall** — governance that placates both sides.

🏭 Pollution

Heavy industry no longer pollutes in the abstract: it **contaminates specific hexes**. Where pollution builds up, the **vegetation around it withers and dies** — and your oxygen falls with it.

> **The answer: Atmospheric Scrubbers** that filter the surrounding hexes. Or move your factories away from your forests.

☣️ The Martian Plague

Now that the planet has **oceans**, something wakes up in them: mutated pathogens Earth has never seen. The infection spreads slowly and your **workforce sickens** — production sags steadily for as long as you ignore it. You get a grace period to prepare. After that, with no medical infrastructure, your output can fall to **half**.

> **The answer:** a **Planetary Isolation Hospital** — but you **must staff it**. An empty hospital cures nobody, and loose doctors with no hospital aren't enough either. **Doctors inside the hospital** — that is the cure.

The Infrastructure Boom

Once the population reaches serious numbers, a second era opens up: an economy on an industrial scale.

💹 The silicon monopoly

Stop selling cheap, raw silicon to Earth. **Process** it instead: the **Quantum Processor Plant** multiplies its value roughly **tenfold**, and the **Interplanetary Stock Exchange** sells off your surplus materials too.

🌋 Drilling into the mantle

The **Deep Core Drill** reaches the planet's mantle and pulls up **limitless** metals — it doesn't even need a deposit. The price: **seismic instability**. The more drills you run, the closer the next **marsquake**, which cracks and disables the buildings around it.

> **The tip: spread them out.** Don't stack drills in one place.

🤖 Drone swarms

The **AI Drone Hive** takes over your heavy industrial buildings: they run at 100% with **no humans at all**, freeing your crew for more important work.

🌀 Extreme weather

Thick air + oceans = **super-storms**. Once enough energy builds up, a **hurricane** breaks out, sweeping away **solar panels** and **flooding** anything built low, knocking them out until they are repaired.

> **The answer: Sea Walls**, which shield the buildings around them.

🐛 Invasive species

Cargo from Earth brings stowaways: **pests**. On a planet full of life they find paradise — they **eat your crops** and **wither your vegetation**. The richer the biosphere, the worse the invasion.

> **The answer:** the **Genetic Vault** and Wildlife Reserves — biodiversity that keeps the balance.

🚄 The Hyperloop network

Your empire has sprawled. The **distant mines** can no longer be supplied properly and run at **half output**. The **Hyperloop Terminal** reconnects them to the colony's core — and you can **chain** terminals, one after another, to reach the far edge of the map.

> **The trap:** if a **hurricane takes down a node**, every mine it was feeding drops **instantly into blackout** — until it is repaired. The network is power, but it is also responsibility.

Sponsors (difficulty levels)

Before you start, you pick a **sponsor**, which sets how easy or hard the run will be:

- **Tech Billionaire** (easy) — plenty of starting capital, mild conditions, rare events.
- **United Space Agency** (normal) — a balanced experience, realistic wear, ordinary event frequency.
- **Private Crowdfunding** (hard) — minimal resources, huge dust storms, fast equipment decay.

The sponsor changes how many supplies you start with, how often disasters hit, and how quickly your equipment wears out.

How a typical run goes

1. **Secure survival** — make sure energy, water, oxygen and food are all in positive flow, so the crew is never at risk.
2. **Grow the economy** — build mines on deposits, stockpile Materials and Credits, and set up labs for research.
3. **Climb the tech tree** — unlock fission power (steady electricity) and industry, so you can afford the big works.
4. **Planetary engineering** — start warming and thickening the atmosphere with gas factories, mirrors and comets. The target gauges begin to climb.
5. **Biosphere** — seed microbes and forests; oxygen rises and the planet turns green.

6. **Victory** — all four targets at 100%. Mars is a living world.

And then the **second act** begins:

7. **The first crisis** — the greenhouse runs away and the migrant ships arrive at the same time. Stabilise climate and housing before it gets away from you.
8. **Society** — you learn to live with factions, pollution and plague; you build governance, scrubbers and hospitals.
9. **The empire** — quantum industry, mantle drilling, drones, a Hyperloop network; your economy becomes a colossus.
10. **Stewardship** — hurricanes, marsquakes and pests never stop. A living planet needs a caretaker, not a conqueror.

There is no single "right" path — you are constantly balancing keeping people alive **now** against investing in the great dream for **later**.

Controls (at a glance)

- **Move the map:** WASD / arrows / mouse drag — **Zoom:** wheel
- **Select a building/tile:** right click
- **Buildings: B** or the icon — **Research: T** or the icon
- **Speed:** Space (pause) and **1 / 2 / 3**
- **Centre on base: H — Next building that needs crew:** `.` — **Exhausted deposit:** `,`
- **Next broken building:** `/` — broken buildings wear a **red wrench** on the map and produce nothing until repaired
- **Crew assignments (drag & drop): C**
- **Reclaim: R** (once unlocked) — **Mute: U**
- **Menu / back: Esc**
- **Help:** the "Help" button in the menu, and the "?" buttons inside the palettes

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Saving, loading & learning

- **Multiple saves** — every save also keeps a **screenshot** of the screen. On the "Load Game" screen you see a list with those images, the date and time of each save; you can enlarge any image, and choose which one to load.
- **Autosave** — the game saves itself every 5 minutes, cycling through 3 slots (Auto 1/2/3).
- **Where the saves live** — in your own account folder, so they survive a rebuild or a reinstall: `%APPDATA%\Mars Terraforming\SavedGames` on Windows, `~/.local/share/MarsTerraforming/SavedGames` on Linux (`$XDG_DATA_HOME` is honoured), `~/Library/Application Support/Mars Terraforming/SavedGames` on macOS. The exact path is written at the bottom of the "Load Game" screen. Saves left next to an older build are moved there automatically at start-up.

- **Tutorial** — from the main menu, a step-by-step guide that teaches you the basics (movement, building, selection, research, speed) as you actually perform them. Leave any time with **Esc**.

In short: you keep a few people alive on a hostile planet, you build an economy, you climb a tech tree, and slowly you turn a frozen rock into a second living world.

And then you discover the hardest part: a living world ***does not belong to you****. It has its own storms, its own diseases, its own politics. Building it was half the game — living with it is the other half.*

That is Mars Terraforming.